SENTATION

Towards a Next-Generation Agricultural Expert System

timodal Conversational Approach

Domain: Artificial Intelligence · Precision Agriculture · NLP
Target: Multi-crop India · Web platform deployment
Supervisor: [Name] | [Department] | [Institution] | 2024–25

146M+
Smallholder farm households in India

Rs.1.5L cr
Annual crop loss from disease & poor advisory

<30%
Farmers with access to agronomist advice

The Agricultural Advisory Crisis in India

Why existing systems fail the 146 million farmers who need them most

India has **1 agricultural extension worker per 1,000+ farmers** — advice is inaccessible at scale (ICAR, 2023)

Crop disease is the **leading cause of yield loss**; most farmers diagnose by eye, often too late (FAO, 2022)

Soil nutrient deficiencies cost **20–30% of potential yield**; soil testing adoption is <15% (NABARD, 2023)

Mandi price information asymmetry causes farmers to sell **30–40% below optimal price** (eNAM, 2023)

Existing AI tools are **English-only, text-only**, or require stable internet — excluding most target users

What AI can offer

→ 24/7 voice advisory in Hindi & regional languages

→ Instant leaf-image disease diagnosis + XAI

→ Soil-sensor-grounded nutrition recommendations

→ Live mandi price advisory with sell/hold guidance

Why now?

→ LLMs (GPT-4o, Gemini 1.5) ready for domain advisory

→ VLMs achieve >90% on PlantVillage disease datasets

→ Bhashini API enables 10+ Indic language ASR/TTS

→ RAG eliminates hallucination in grounded advisory

Landscape of Prior Research

Six research streams this thesis builds upon and extends

S1 · Multimodal conversational

MIRAGE (2025) · AgMMU · Farmer.Chat (ACM 2024)
→ Benchmarks & systems; lack unified advisory session

S2 · LLMs & RAG for farm ad

AgroLLM · AgriLLM · J. Agri. Engineering RAG (2026)
→ Strong text-only RAG accuracy; no image or voice modality

S3 · VLMs for plant disease

LLMI-CDP (Sci. Reports 2025) · AgriChain · Self-consistency VLMs
→ 91–94% leaf accuracy; no farmer-facing conversational UI

S4 · Explainable AI in agricul

XAI-CROP (Springer NCA 2024) · Grad-CAM plant disease
→ XAI per task in isolation; never surfaced to farmers conversationally

S5 · Speech & multilingual NLP

Indic ASR benchmark (2025) · IndicBERT · Bhashini
→ Voice interfaces exist; not connected to advisory backend

S6 · Agentic & expert system

CropiBot (Sci. Reports 2026) · Transformer survey (EAAI 2024)
→ Single-agent chatbots; lack specialist routing & sensor fusion

Key papers cited

- [1]
- [3]
- [4]
- [5]
- [6]
- [7]
- [9]
- [10]
- [8]
- [12]

What No Existing System Has Built

Five specific gaps identified across the literature survey

G1 No unified multimodal advisory session

No system combines disease (image) + nutrition (sensor) + crop rec + weather + live market data in one conversational turn

G2 No cross-modal sensor–image nutrition fusion in a dialogue

Sensor fusion papers work offline; no conversational interface wraps leaf image + soil NPK data for farmer-facing nutrition advice

G3 XAI not rendered to the farmer in any deployed system

XAI-CROP [7] and LLMI-CDP [5] generate SHAP/Grad-CAM as research output — never surfaced as a visual explanation in a farmer UI

G4 No end-to-end Indic voice-to-advisory pipeline

Indic ASR benchmark [10] achieves WER 16% for Hindi but no system connects it to an agronomic RAG backend with TTS response

G5 Live mandi price never co-delivered with agronomic advice

Market agents and agronomic agents exist in silos across all surveyed literature — no system answers both in the same conversational response

Formal Problem Statement

Indian smallholder farmers face **simultaneous, interdependent challenges** — crop disease, nutrient deficiency, uncertain crop selection, irrigation scheduling, and price realisation — that no single AI model or modality can address. Existing expert systems are **text-only, single-agent, single-language, and non-explainable**, making them inaccessible and untrustworthy for the target population. **No current system** accepts voice in Hindi, diagnoses disease from leaf images, integrates soil sensor data, and advises on live mandi prices within a single conversational turn — with explanations grounded in verified agronomic knowledge sources.

Challenges to overcome

Modality heterogeneity — voice, image, text, sensor, price are structurally different inputs

Domain depth — agriculture requires expert-level, crop- and region-specific knowledge

Hallucination risk — LLMs fabricate crop-care advice with dangerous confidence

Language barrier — Indic ASR accuracy sub-optimal for agricultural vocabulary

Trust — farmers reject advice they cannot understand or verify

Solution constraints

Must work on commodity Android device + 3G connection

Response time <4 s (cloud) · <8 s (PWA offline)

Knowledge grounded exclusively in ICAR, FAO, CGIAR verified sources

Explainability must be farmer-readable — visual, not numerical

Multi-crop: wheat, rice, maize, cotton, tomato, potato and more

Research Objectives

Eight measurable objectives mapping directly to identified research gaps

O1 · Architecture Design LangGraph multi-agent orchestrator: intent classifier, parallel dispatch, Redis multi-turn memory

O2 · Disease Fine-tune ViT-L/16 + LoRA on PlantVillage/PlantDoc/IP102 — >90% top-1, 200+ diseases, 50+ crops

O3 · Nutrition Cross-modal MLP fusing ResNet-50 leaf features with NPK/pH/EC sensor readings for fertiliser dose

O4 · RAG Hallucination-resistant RAG: 250k+ ICAR/FAO chunks in Qdrant, hybrid BM25+dense, LLM-as-judge gating

O5 · XAI Surface Grad-CAM heatmaps & SHAP bar charts in React PWA as farmer-facing UI — not research output

O6 · Voice Whisper large-v3 (Indic fine-tuned) ASR → Bhashini TTS — WER <20% Hindi, <30% regional, <4 s latency

O7 · Market Agmarknet + eNAM live pull (15-min), Prophet 7-day forecast, sell/hold advisory co-delivered with agronomy

O8 · Evaluation Benchmark on MIRAGE, AgMMU; validate with 10 ICAR-certified agronomists on 200 real farmer queries

System Approach — Six-Layer Architecture

Each layer addresses specific research objectives and closes identified gaps

L1 · Input layer Text · Voice (Whisper ASR) · Image (leaf/field) · Sensor (NPK/pH/EC) · Live price (Agmarknet/eNAM) · 10 Indic languages via Bhashini

L2 · Preprocessing Parallel: ASR → Hindi norm · Image QA (BRISQUE) + 448×448 resize · Sensor calibration · Price anomaly filter

L3 · Orchestrator GPT-4o backbone + fine-tuned BERT intent classifier · LangGraph dispatch · Redis session memory · Confidence-gated response merge

L4 · Specialist agents ×5 **Disease:** ViT-L LoRA+Grad-CAM | **Nutrition:** ResNet-50+sensor MLP+SHAP | **Crop:** XGBoost+SHAP | **Weather:** LSTM+IMD | **Market:** Agmarknet+Prophet

L5 · Knowledge + RAG E5-large embeddings · Qdrant · Hybrid BM25+dense · Cross-encoder rerank · HyDE · Qwen3 LLM-as-judge · 250k+ ICAR/FAO/CGIAR chunks

L6 · Output + XAI React 18 PWA · Grad-CAM overlay · SHAP bar charts · Bhashini TTS Hindi/regional · District map · Price trend chart · FastAPI WebSocket

Evaluation Framework

Quantitative metrics per component — each tied to a research objective

Disease diagnostic (O2)

Dataset: PlantVillage test + PlantDoc
Metrics: Top-1 Acc · F1 macro · AUC-ROC
Target: >90% top-1 · 200+ diseases
XAI: Grad-CAM IoU vs. expert annotation

Crop recommendation (O4)

Dataset: ICRISAT district yield + IMD climate
Metrics: Top-3 Acc · Agronomist agreement %
Target: Top-3 acc >85% · agreement >80%
XAI: SHAP feature rank correlation

Nutrition agent (O3)

Dataset: ICAR soil health card DB + synthetic sensor
Metrics: RMSE fertiliser dose · F1 deficiency class
Target: RMSE <5 kg/ha · F1 >0.85
XAI: SHAP stability score across runs

ASR / Voice pipeline (O6)

Dataset: Indic ASR benchmark (10,934 recordings)
Metric: WER per language
Target: WER <20% Hindi · <30% Indic
E2E latency: voice → response <4 s

RAG quality (O4)

Framework: RAGAS evaluation suite
Metrics: Faithfulness · Context relevance · Answer rel.
Target: Faithfulness >0.88 · Hallucination <5%
Benchmarks: MIRAGE · AgMMU eval sets

Expert agronomist panel (O8)

Panel: 10 ICAR-certified agronomists
Metric: 5-point Likert correctness + clarity
Target: Mean correctness >4.0 / 5.0
Blind eval: 200 real farmer queries

Ablation studies: 5-agent vs. single-agent baseline · RAG vs. no-RAG · XAI rendered vs. not · voice vs. text-only

Five novel contributions over prior art and a 12-month delivery plan

C1 First multi-agent system routing to five specialist domain models

Closes G1 · novel over: Farmer.Chat, AgroLLM, CropiBot (all single-model)

C2 Cross-modal leaf image + soil sensor fusion for nutrition in a dialogue

Closes G2 · novel over: all surveyed nutrition papers (offline inference only)

C3 Farmer-facing Grad-CAM heatmaps and SHAP charts rendered in web UI

Closes G3 · novel over: XAI-CROP [7], LLMI-CDP [5] — both log XAI internally

C4 End-to-end Indic voice pipeline: Whisper ASR → Hindi RAG → Bhashini TTS

Closes G4 · novel over: Indic ASR benchmark [10] paper (no advisory backend)

C5 Live mandi price co-delivered with agronomic advice in one conversational turn

Closes G5 · novel over: all surveyed papers — no prior system attempts this integration

M1–3	M4–6	M7–9	M10–12
Data collection · KB · RAG	Training · Fusion · ASR	Orchestrator · PWA · QA	Eval · Panel · Thesis

References

Key papers cited in this research

- MIRAGE: A Benchmark for Multimodal Information-Seeking and Reasoning in Agricultural Expert-Guided Conversations. arXiv, 2025.
- A Multimodal Conversational Assistant for Agricultural Plots from Geospatial Open Data. arXiv, 2025.
- AgMMU: Agricultural Multimodal Understanding and Reasoning Benchmark. arXiv, April 2025.
- Farmer.Chat: Scaling AI-Powered Agricultural Services for Smallholder Farmers. ACM, 2024. arXiv:2409.08916.
- LLMI-CDP: A Large Language Model for Multimodal Identification of Crop Diseases and Pests. Scientific Reports (Nature), 2025.
- AgriChain: Visually-Grounded Expert-Verified Reasoning for Agricultural VLMs. arXiv, 2025.
- Enhancing Crop Recommendation with XAI (XAI-CROP). Neural Computing and Applications, Springer, 2024.
- RAG-Based LLM Advisory Framework for Farmers. Journal of Agricultural Engineering, 2026.
- Agentic AI-Driven Autonomous Decision Support System for Smart Agriculture (CropiBot). Scientific Reports, 2026.

References

Key papers cited in this research

- Benchmarking ASR for Indian Languages in Agricultural Contexts. arXiv, 2025.
- Harnessing Large Vision and Language Models in Agriculture: A Review. Frontiers in Plant Science, 2025.
- Recent Advances in Transformer Technology for Agriculture: A Survey. Engineering Applications of AI, Elsevier, 2024.
- Self-Consistency in VLMs for Precision Agriculture. arXiv, 2025.
- Overcoming LLM Challenges using RAG-Driven Precision in Coffee Leaf Disease Remediation. IEEE Conference, 2024.
- Deep Learning and Computer Vision in Plant Disease Detection. Artificial Intelligence Review, Springer, 2025.